

Session 7: More data visualisation and ggplot

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2026-06-08



Overview I

1 Introduction

2 Additional variable in ggplot()

3 Wrapping up

4 Working on projects



Learning goals of today

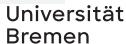
- Short repetition from last week
- Some more data viz examples

Additional variable in ggplot()



Options for additional variable

- There are many ways to include additional variable into graph
 - *colour / fill*
 - *size*
 - *shape*
 - *alpha*



colour (continuous): Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with colour (continuous)
ggplot(data = iris,
        mapping =
          aes(x = Sepal.Length,
              y = Sepal.Width,
              colour = Petal.Width)) +
  geom_point()
```



size: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with size
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width,
            size = Petal.Width)) +
  geom_point()
```





shape

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- Can be useful to visually highlight different sub-divisions in data

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- Can be useful to visually highlight different sub-divisions in data
- Originally more useful for black and white applications, nowadays colours used more regularly

shape: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width,
            shape = Species)) +
  geom_point()
```



Combining options *colour* and *shape*

- Of course, one can also combine options

Combining options *colour* and *shape*

- Of course, one can also combine options
- may increase visibility of differences even more

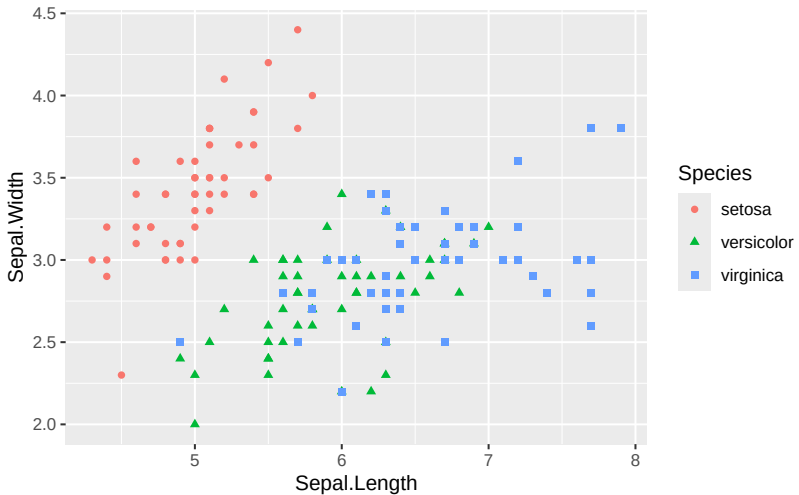
colour + shape: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with colour and shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
             y = Sepal.Width,
             colour = Species,
             shape = Species)) +
  geom_point()
```



colour + shape: Example image



alpha

- *alpha* sets the opacity of information

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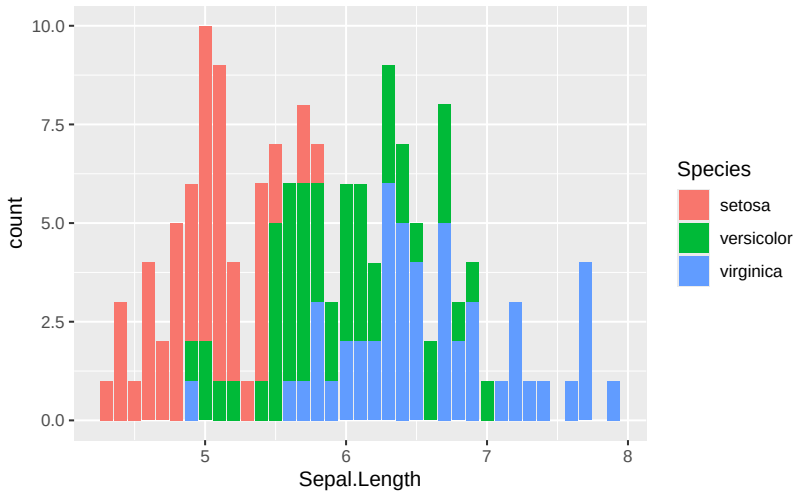
- *alpha* sets the opacity of information
- Can be useful to visually highlight different sub-divisions in data
- Originally more useful for black and white applications, nowadays colours used more regularly
- Remember the bar plot from before?

fill (factor): Example code

```
# Load ggplot2
library(ggplot2)

# Bar plot using iris data set with fill
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            fill = Species)) +
  geom_bar()
```

fill (factor): Example image



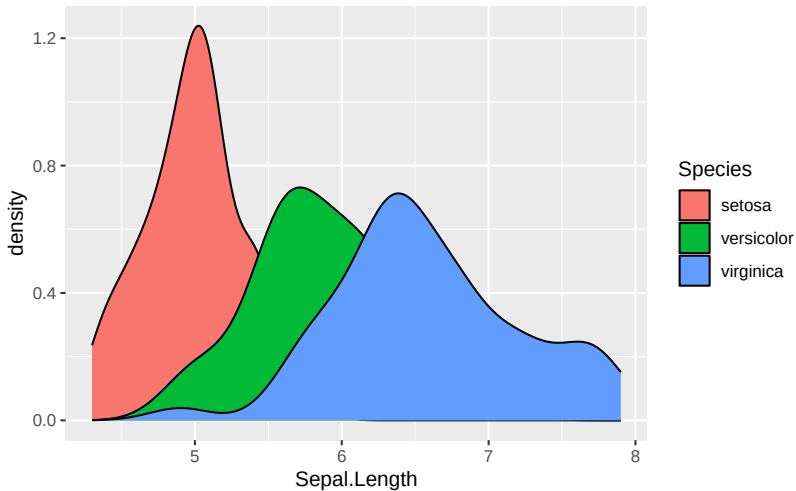
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fill (density): Example code

```
# Load ggplot2
library(ggplot2)

# Density plot using iris data set with fill
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            fill = Species)) +
  geom_density()
```

fill (density): Example image



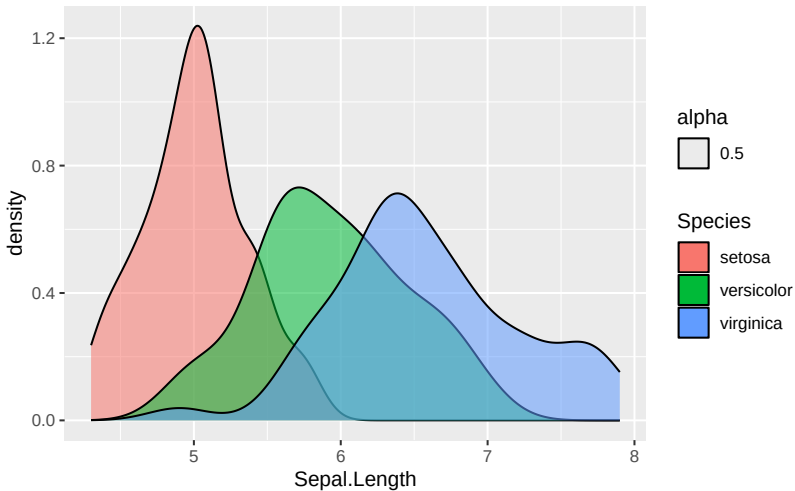
fill (density) + alpha: Example code

```
# Load ggplot2
library(ggplot2)

# Density plot using iris data set with fill and alpha
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
             fill = Species,
             position = "identity",
             alpha = 0.5)) +
  geom_density()
```



fill(density) + alpha: Example image



alpha as variable

- Of course, *alpha* can also be used with variable

alpha as variable

- Of course, *alpha* can also be used with variable
- Though use case less obvious

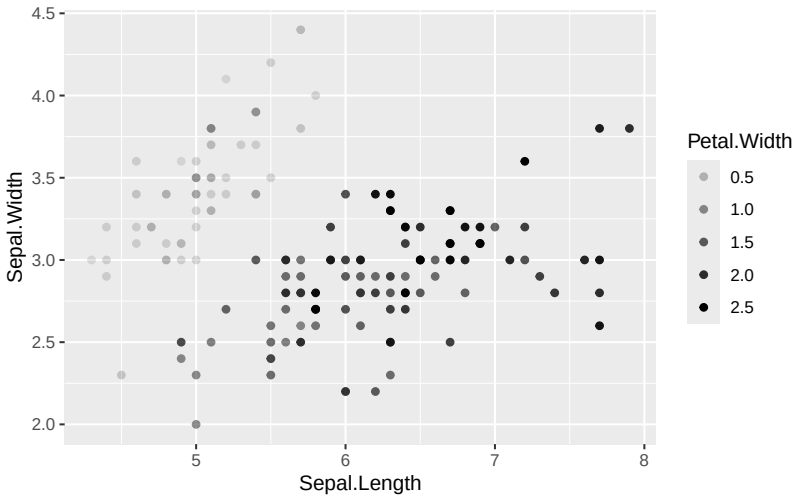
alpha: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with shapes
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width,
            alpha = Petal.Width)) +
  geom_point()
```



alpha: Example image

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Exercise: Create three graphs with your data

- Create one graph using *size*

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Exercise: Create three graphs with your data

- Create one graph using *size*
- Create one graph using *shape* and *colour*
- Create a density plot with *fill* and *alpha*
- Consider using different variables than in the last exercise

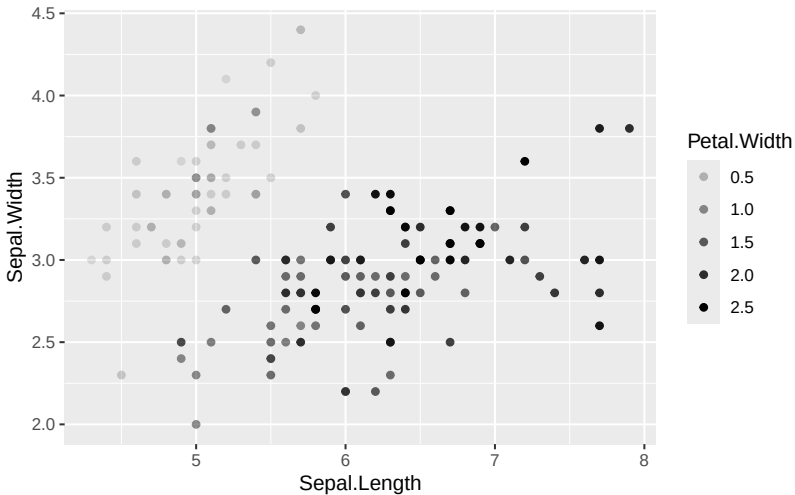
alpha: Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with size
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width,
            alpha = Petal.Width)) +
  geom_point()
```



alpha: Example image

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Great example: Faceting

The number of Europeans travelling to the US has cratered under Trump

Year-on-year change in visitors to the US, by country of origin (%)



Source: [International Trade Administration, U.S. Department of Commerce](#)
FT graphic: John Burn-Murdoch / @jburnmurdoch
©FT

facet_wrap() / *face_grid()*

- faceting is super useful, especially with large data sets

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facet_wrap() / *face_grid()*

- faceting is super useful, especially with large data sets
- Creates separate graph per sub-group (hence requires factor variable)
- *facet_wrap(~ x)* <- best to use with one factor variable (x), plots separate graphs per factor
- *facet_grid(x ~ y)* <- best to use with two factor variables (x, y), plots grid of all combinations of x and y

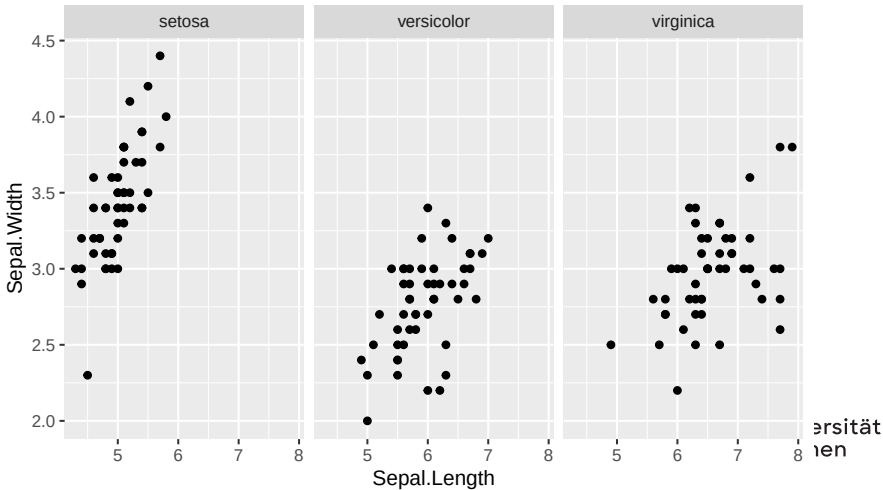
facet_wrap(): Example code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set with facet_wrap()
ggplot(data = iris,
       mapping =
         aes(x = Sepal.Length,
            y = Sepal.Width)) +
  geom_point() +
  facet_wrap(~ Species)
```



facet_wrap(): Example image



Exercise: Create two graphs with your data

- Create one graph using `facet_wrap`

Exercise: Create two graphs with your data

- Create one graph using `facet_wrap`
- Create one graph using `facet_grid()`

- Which topics would you like me to still cover (more)?

Working on projects